

CMPT 742: Practices in visual computing I

Assignment 2

Image segmentation

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1. Default Parameters + Horizontal/Vertical flip

Data augmentation strategies: Horizontal/Vertical flip

Size of the images: 572

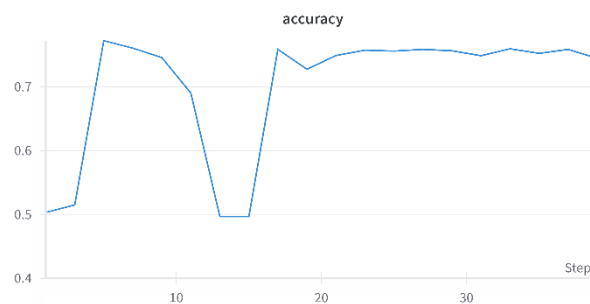
Number of Epochs: 20

Batch size: 4

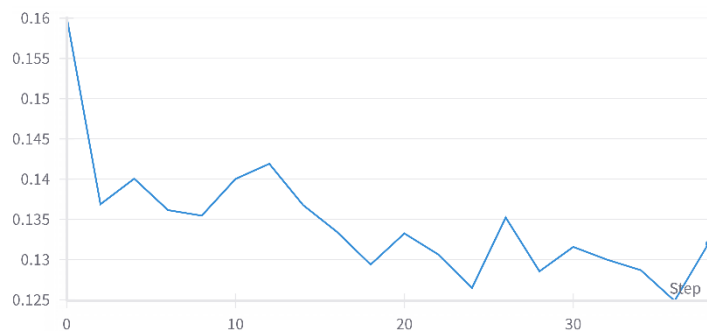
Learning rate: 0.01

Times taken: 2m 27s

Final accuracy: 0.74458

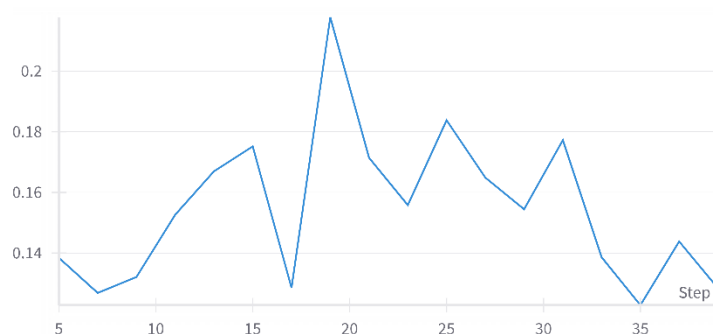


Epoch_Loss



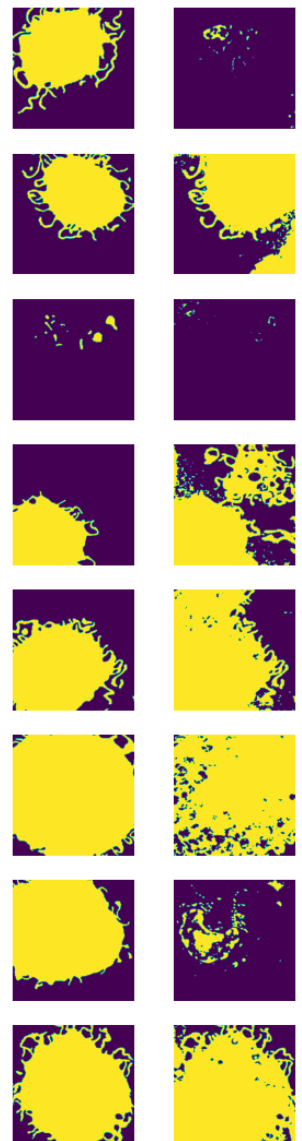
Train loss:

validation_loss



Test loss:

Result:



2. Default Parameters + Rotation

Data augmentation strategies: **Rotation**

Size of the images: 572

Number of Epochs: 20

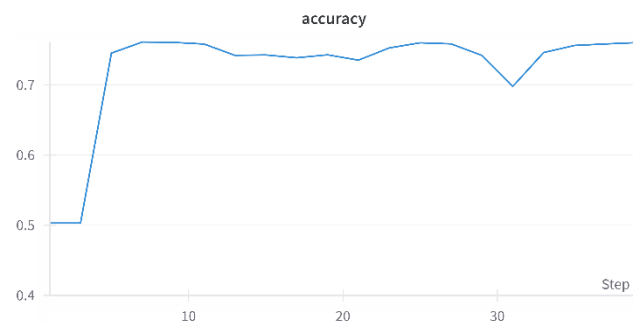
Batch size: 4

Learning rate: 0.01

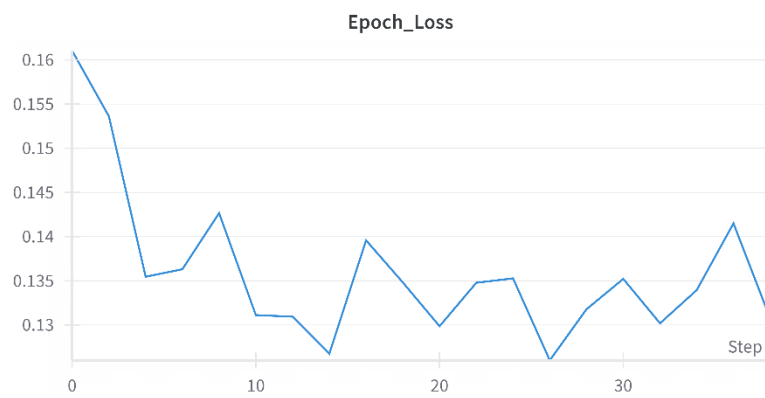
Times taken: 1m 47s

Final accuracy: 0.76051

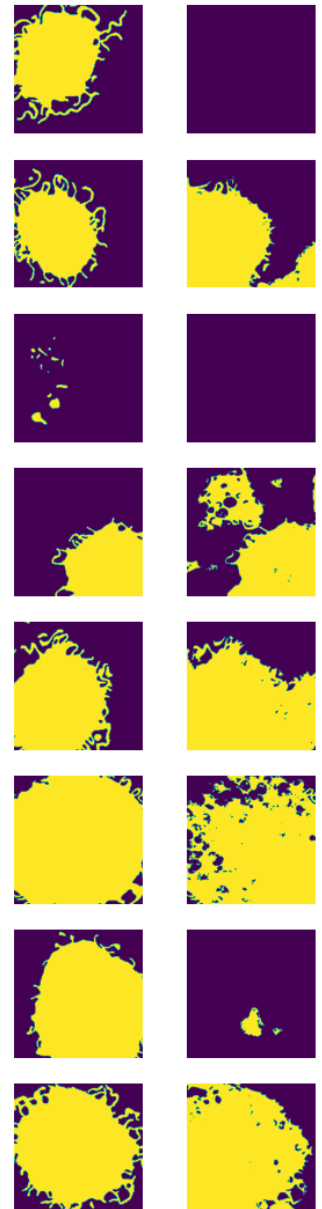
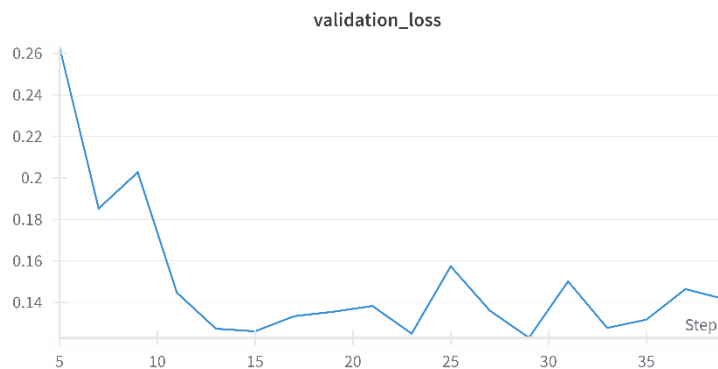
Result:



Train loss:



Test loss:



3. Default Parameters + Gamma correction

Data augmentation strategies: **Gamma correction**

Size of the images: 572

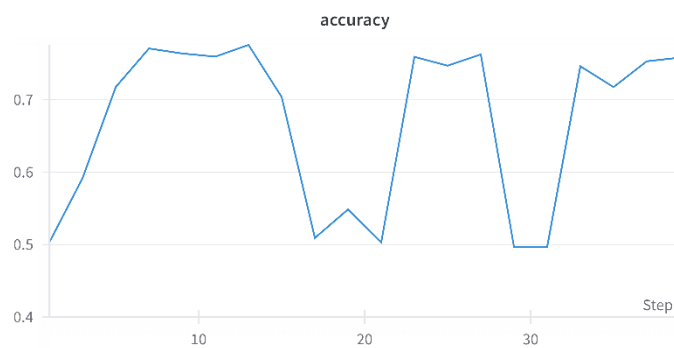
Number of Epochs: 20

Batch size: 4

Learning rate: 0.01

Times taken: 2m 46s

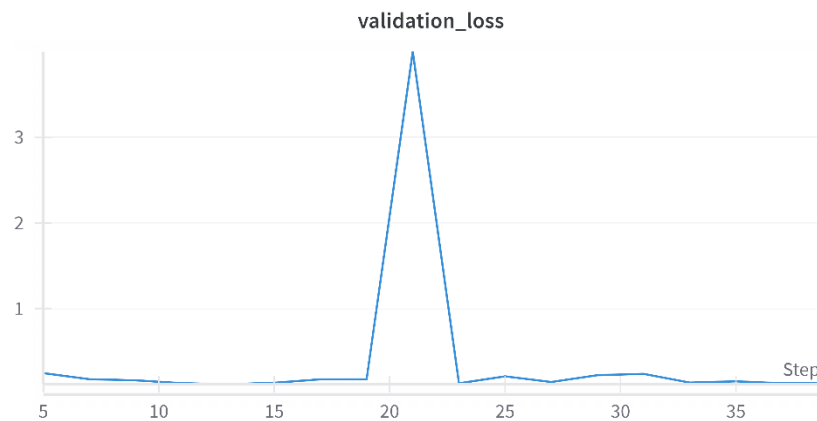
Final accuracy: 0.75828



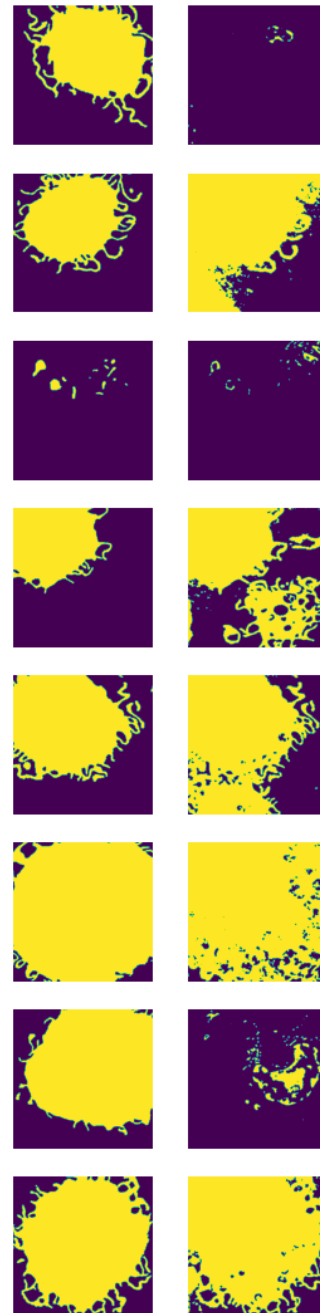
Train loss:



Test loss:



Result:



4. Default Parameters + Elastic Transform

Data augmentation strategies: Elastic Transform

Size of the images: 572

Number of Epochs: 20

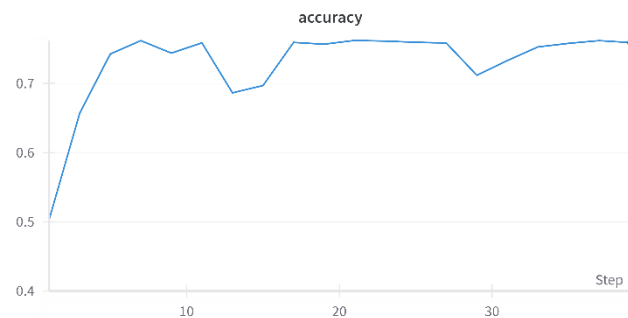
Batch size: 4

Learning rate: 0.01

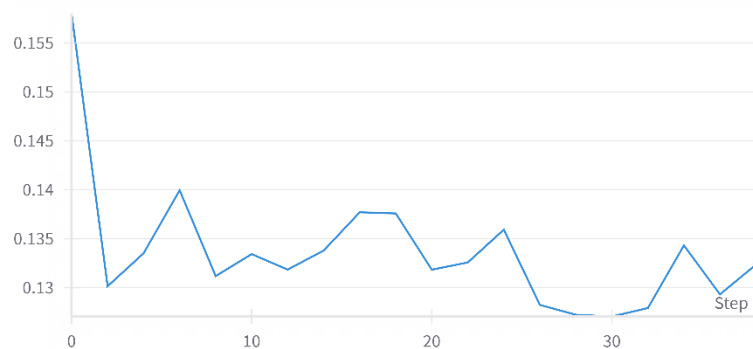
Times taken: 3m 31s

Final accuracy: 0.75897

Result:

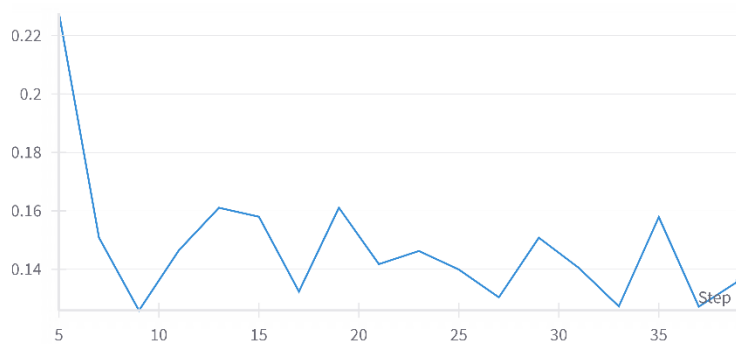


Epoch_Loss

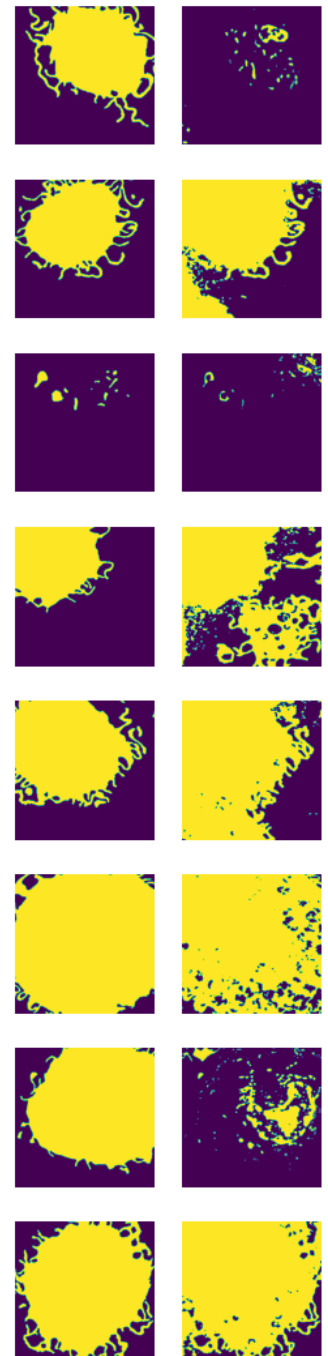


Train loss:

validation_loss



Test loss:



5. Modified Parameters + Random augmentation

Data augmentation strategies: **Random augmentation from all above**

Size of the images: 572

Number of Epochs: **25**

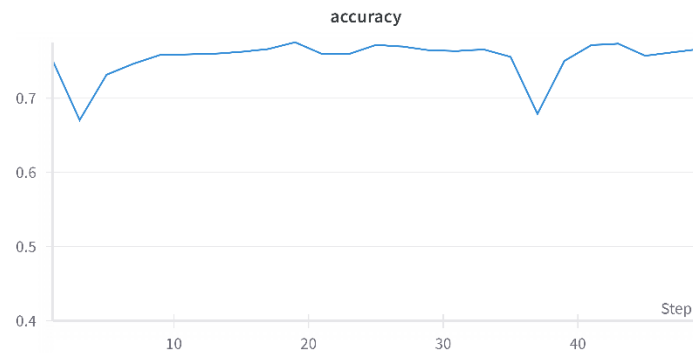
Batch size: **2**

Learning rate: **$1e-5$**

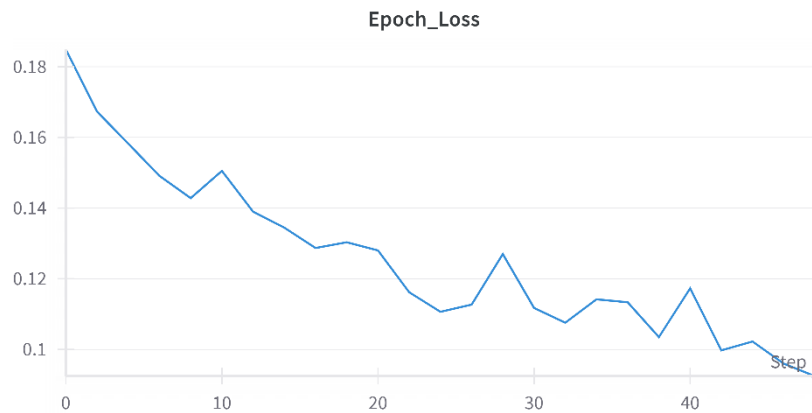
Weight decay = **$1e-7$**

Times taken: 2m 54s

Final accuracy:
0.76612

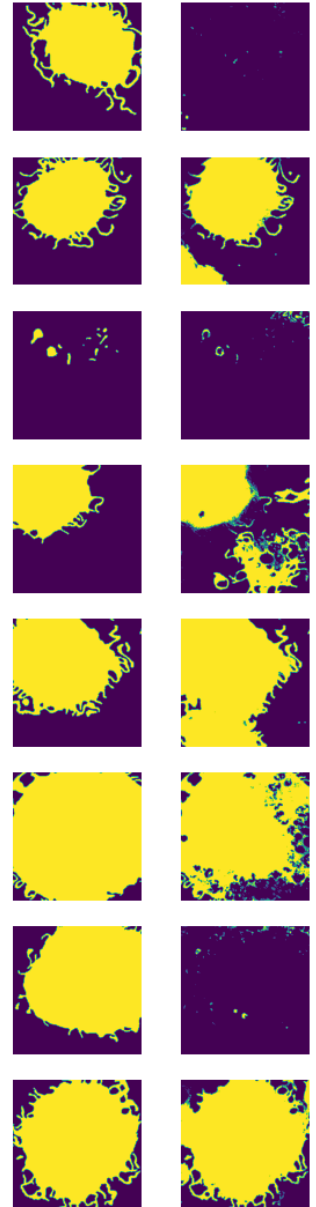


Train loss:



Test loss:

Result:



6. Modified Parameters + Random augmentation

Data augmentation strategies: **Random augmentation from all above**

Size of the images: 572

Number of Epochs: **30**

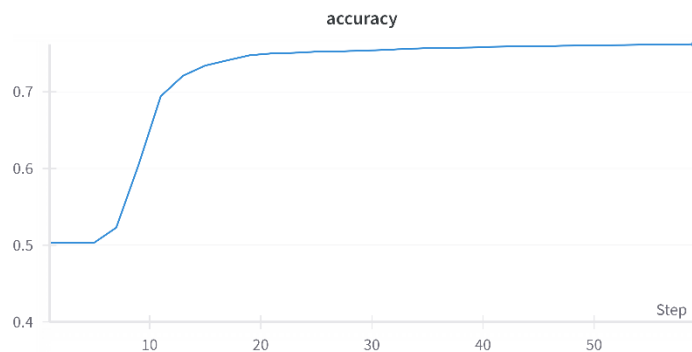
Batch size: **2**

Learning rate: **$1e-6$**

Weight decay = **$1e-5$**

Times taken: 4m 16s

Final accuracy: 0.76204

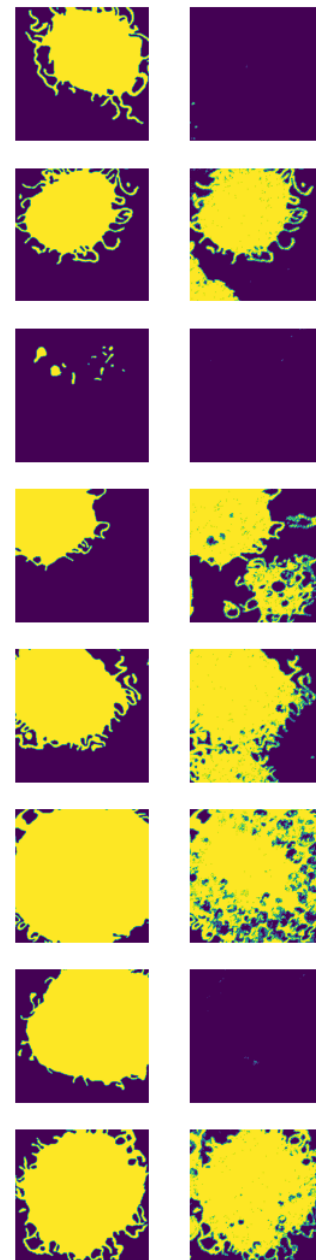


Train loss:

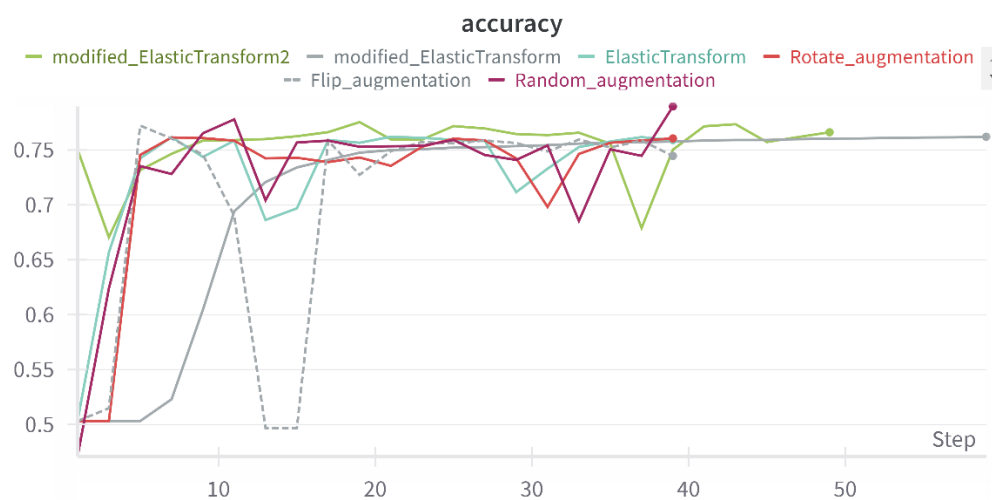
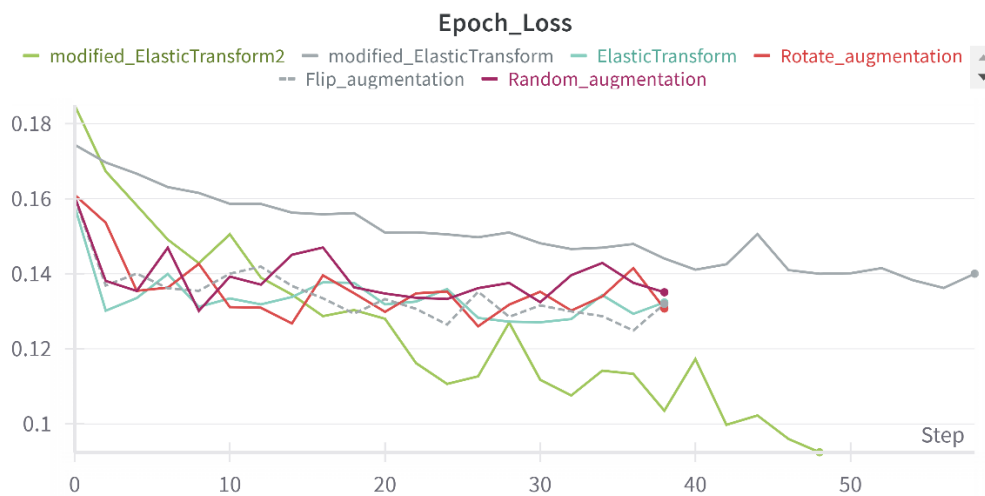
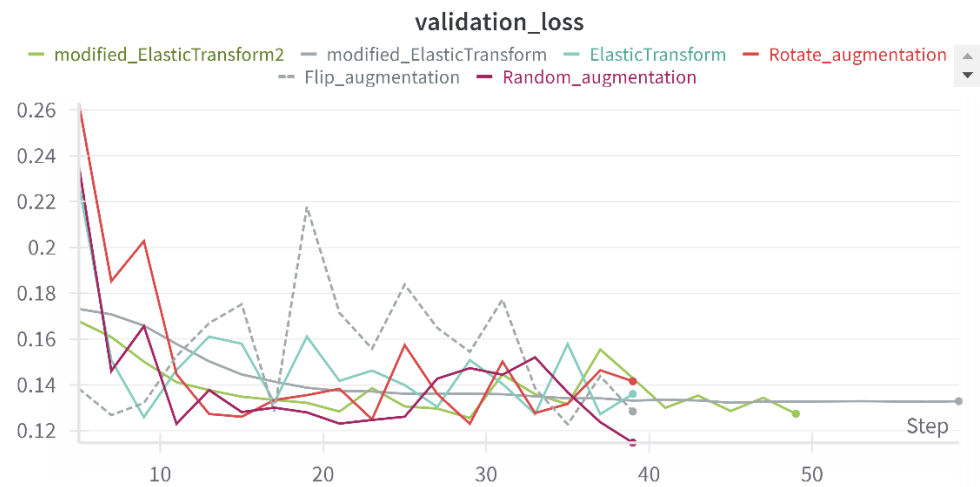


Test loss:

Result:

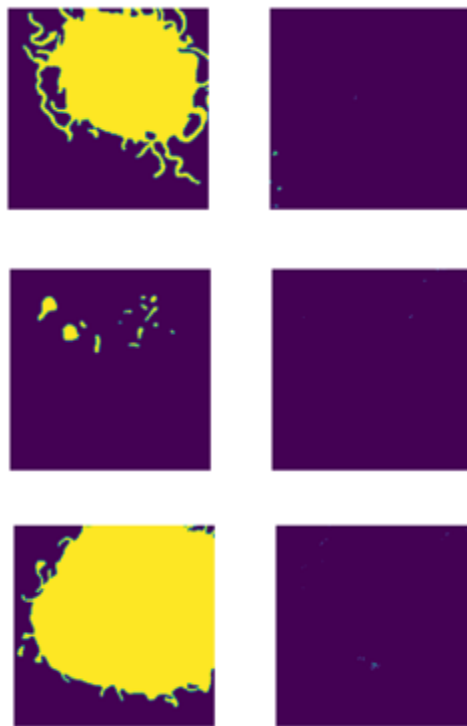


Compare all results with weights and biases:

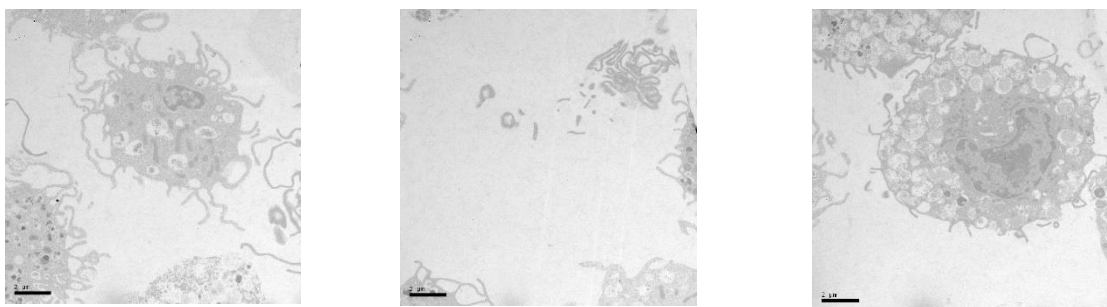


All above is done by Oct. 27, and I tried different methods including changing he parameters, modified the data augmentation strategies, and using different optimizer (Adam and SGD with scheduler, and their combination), and some deviate from the original architecture, but the accuracy still stays around 76-78%.

On the last day, I looked at the output test images carefully. The errors usually occurred in the first, third, and seventh images, which got blank masks.



I originally thought that this was because the model was overfitting in the training set, but when I looked at the corresponding images in the test set, they had one thing in common: lighter colors/low grayscale values.



So, I performed data preprocessing (histogram equalization) in the data loader, which greatly improved the accuracy, and the final result was:

7. Modified Parameters + Random augmentation + Histogram

Data augmentation strategies: **Random augmentation from all above**

Size of the images: 572

Number of Epochs: **25**

Batch size: **2**

Learning rate: **1e-6**

Adam weight decay = **1e-5**

SGD weight decay = **1e-4**

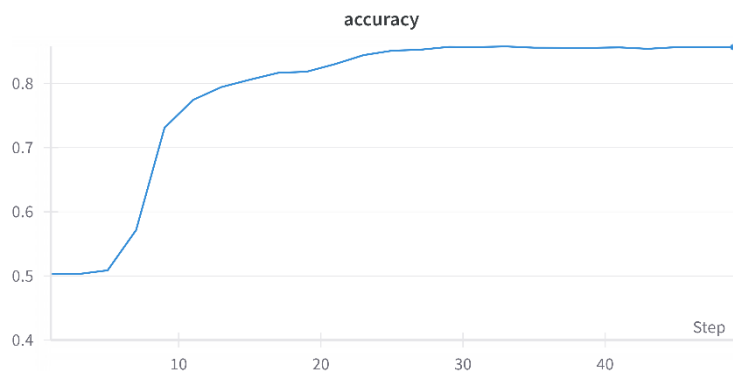
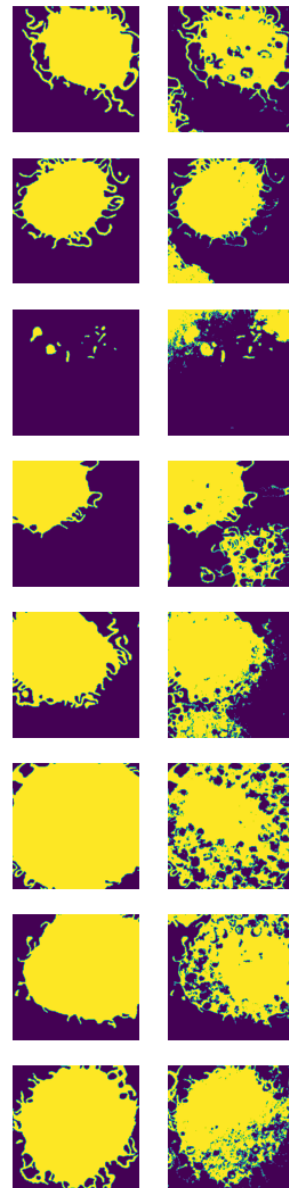
Scheduler = StepLR(step_size=10, gamma=0.9)

Dropout rate= 0.2

Times taken: 2m 47s

Final accuracy: 0.85663

Result:



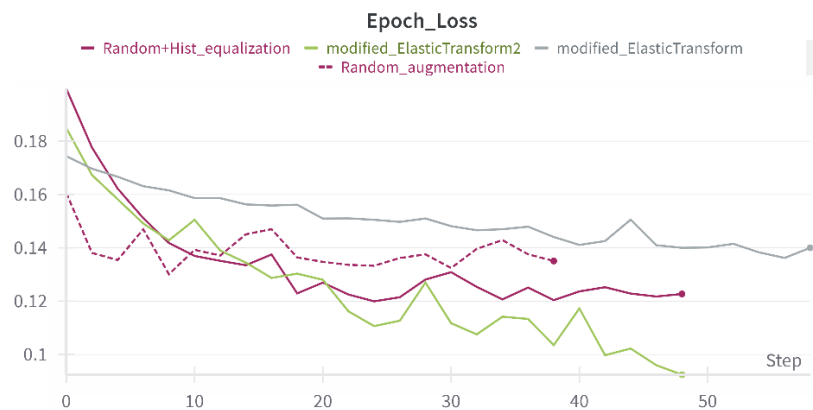
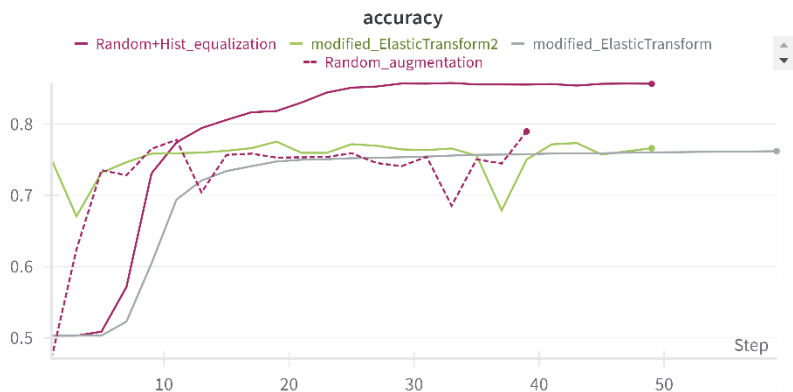
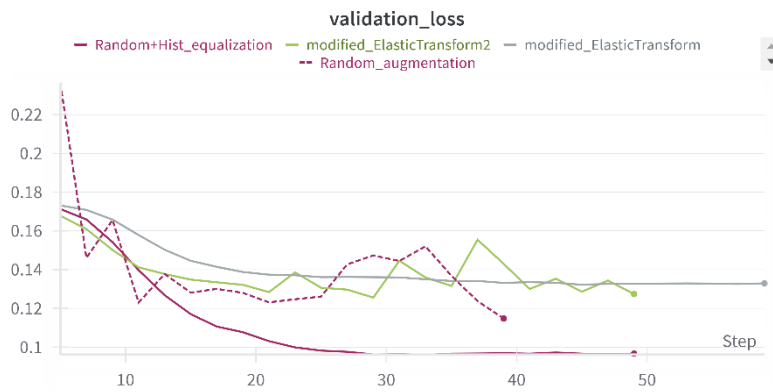
Train loss:



Test loss:



And again, compare it to the best 3 results with weights and biases:



Obviously, after adding the preprocessed, the model got better performance on the accuracy and test loss.